

Analytical Appendix

Query results, method notes and validation design behind the End-to-End Case Study.

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A	Query map
B	Baseline results
C	Category 46 investigation results
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All results below are read from incidents_final_v2 (the DQ-08 corrected model) unless marked "v1, superseded". Every figure marked *run to populate* in an earlier draft of this appendix has now been run; none remain.

Where this analysis comes from

This appendix supports an independent case study whose origin is an internship observation at Protiviti: incidents resolved one at a time, with no record or feedback loop. The analysis itself runs on a separate, real, anonymised, publicly available ServiceNow dataset, not internship or client data. See *02_End_to_End_Case_Study.pdf*, section 2, for the full framing.

A. Query map

File	Query	Purpose	Key result
01	P1, P2	Rows vs distinct incidents	119,998 rows; 20,769 incidents
01	P3, P4	Rows per incident	Max 58 (INC0019396); most common is 3 rows (4,942)
01	P5	Incident states	Two terminal states; artefact value -100

File	Query	Purpose	Key result
01	P6, P7	Date format; placeholders	Day-first text; ? on cmdb_ci (20,717), assignment_group (2,157), category (7)
02	v1 view	One row per incident (superseded)	20,769 rows (22,136 before tie-breaker); text-sort bug, DQ-08
02	v2 view + diagnostic	Corrected ordering (DQ-08)	Run and reconciled: 8,018 incidents (38.6%) selected a different row than v1
03	B1 to B4	Volume, resolution, SLA by group	See section B
03	B5, B6	Overall breach; backlog	39.4% (8,174 of 20,769); 1 flagged, likely not open
04	R1, R2	Asset recurrence attempt	Rejected: 99.75% missing
04	R3, R4	Caller coverage; pair count	73,359 pairs at 14 days (v2)
04	R5, R6	Caller + category summary; time span	See section D
05	C1 to C8	Candidate evidence and KPI baselines	See section C
06	V1 to V4	Windows, incident-level flags, robustness	Run and reconciled (section E)

B. Baseline results

B1. Incidents by category (top 15)

Category	Incidents	Category	Incidents	Category	Incidents
Category 26	2,823	Category 9	940	Category 61	698
Category 42	2,814	Category 37	936	Category 24	547
Category 53	2,345	Category 23	879	Category 45	539
Category 46	1,993	Category 20	869	Category 51	525
Category 32	1,276	Category 57	731	Category 28	491

B3. Average resolution time, categories with 40+ incidents (corrected model)

Category	Avg days	Incidents
Category 34	57.6	439
Category 55	28.1	106
Category 22	22.1	47
Category 45	19.9	526
Category 46	13.9	1,919
Category 19	12.1	199
Category 24	11.6	541
Category 61	7.8	696

Excluded from this reading: the ? placeholder and categories below 40 incidents. Resolution time is unaffected by the DQ-08 correction (opened_at/resolved_at do not change between v1 and v2 row selection); Category 34 is about 58 days on both models (57.8 vs 57.6).

B4. SLA breach rate by assignment group, groups with 100+ incidents — corrected model

Group	Incidents	Breached	Breach rate
Group 10	271	232	85.6%
Group 37	133	105	78.9%
Group 31	168	124	73.8%
Group 66	334	231	69.2%
Group 72	298	205	68.8%
Group 57	254	172	67.7%
Group 22	141	94	66.7%
Group 29	200	133	66.5%
Group 33	172	110	64.0%
Group 76	165	102	61.8%
? (unassigned)	2,157	1,108	51.4%
Group 70 (benchmark)	7,369	1,373	18.6%
All incidents	20,769	8,174	39.4%

v1 (superseded) figures for comparison: Group 5 32.5%, Group 48 32.4%, Group 66 31.7%, Group 12 31.1%, Group 37 30.7%, Group 31 29.0%; Group 70 benchmark 8.4% (of 7,189); all incidents 16.9% (3,517 of 20,769). See Data_Quality_Log.md, DQ-08.

C. Category 46 investigation results (query C1) — corrected model

Group	Incidents	Avg reassign.	Breached	Breach rate
Group 70	294	0.4762	73	24.8%
Group 57	198	0.9596	130	65.7%
Group 6	175	1.5657	101	57.7%
Group 39	169	1.0000	60	35.5%
Group 10	151	1.3841	115	76.2%
Group 37	119	0.8067	93	78.2%
? (unassigned)	108	0.7222	29	26.9%
Group 64	101	0.0495	43	42.6%

Eight of 38 shown (37 named + unassigned). Total: 1,993 incidents; the largest (Group 70) holds 294, 14.8%. Groups with fewer than about 10 incidents show breach rates that carry no statistical weight.

Baselines (C5, C7a, C7b, C8) — corrected model

Metric	Value
Category 46 average reassignments	0.9237
Dataset-wide average reassignments (comparator, C7a)	0.9880
Category 46 rank by avg. reassignments (C7b, n≥40)	18th of 26 — not high, in line with the dataset average
Category 46 SLA breach rate	54.1% (dataset-wide 39.4%)
Category 46 incidents from callers with 3+ raises (C8)	1,079 of 1,993 (54.1%), from 203 such callers
Caller 1904 / Category 51 incident count	399 (all Group 64; reassignments 0.0000; reopens 0.0000; 22 breached)

v1 (superseded): avg. reassignments 0.8343, breach rate 25.9% against a 16.9% dataset-wide rate. The C7a/C7b comparator run shows Category 46's reassignment rate was never actually elevated at any point — on either model it sits close to or slightly below average. Only the breach rate is a real, corrected signal.

D. Recurrence results

D1. Caller coverage (R3) and pair count (R4)

Caller	Incidents	Caller	Incidents
Caller 1904	437	Caller 4414	52
Caller 290	239	Caller 3763	49
Caller 4514	128	Caller 90	49

Pairs (same caller, same category, second incident 0 to 14 days after the first), corrected model: **73,359**. (v1, superseded: 73,339.)

D2. Caller + category combinations seen 3 or more times (R5, top 20)

Caller	Category	Incidents	Caller	Category	Incidents
Caller 1904	Category 51	399	Caller 4166	Category 42	19
Caller 290	Category 28	212	Caller 4414	Category 45	18
Caller 4514	Category 28	47	Caller 1531	Category 46	18
Caller 4514	Category 23	42	Caller 90	Category 28	17
Caller 3763	Category 46	28	Caller 93	Category 46	17
Caller 1904	Category 46	26	Caller 116	Category 42	16
Caller 1717	Category 37	23	Caller 92	Category 57	15
Caller 2630	Category 46	23	Caller 4890	Category 23	15
Caller 1441	Category 46	20	Caller 4414	Category 46	14
Caller 290	Category 42	20	Caller 2735	Category 46	14

D3. Time span of the two extreme callers (R6, dates parsed correctly)

Caller	First seen	Last seen	Incidents
Caller 1904	2016-03-05 10:07	2016-05-09 06:24	437
Caller 290	2016-03-09 18:39	2016-05-12 22:20	239

The first attempt compared the dates as text and returned Caller 290's first date after its last. Parsing with STR_TO_DATE before MIN/MAX fixed it — the same bug class as DQ-08, caught independently in this query (logged as DQ-07).

E. Sensitivity and reconciliation — run and reconciled

The framework asks for the recurrence window to be tested, not assumed. Two independent implementations of the same rule exist (SQL and Python), so that logic errors in either one show up as a mismatch. Both have now been run against the real file.

Window	Pairs (SQL, V1)	Pairs (Python)	Incidents flagged (V2)	Pairs excl. Callers 1904, 290 (V3)
7 days	51,007	51,007 (PASS)	4,213 (20.3%)	7,291
14 days	73,339	73,339 (PASS)	5,238 (25.2%)	9,647
30 days	100,642	100,642 (PASS)	6,406 (30.8%)	13,322

Flagged incidents and V3 pairs are on the v2 (corrected) rows and pair with V4 (51,014 / 73,359 / 100,701). The V1 pair columns are the earlier run, kept as a note; v1 flags were 4,210, 5,228 and 6,380 (superseded).

Definition (identical in both implementations)

Same caller_id and same category; caller not blank. The later incident opens 0 to W days after the earlier one, using the calendar-day difference (MySQL DATEDIFF on parsed dates). Each pair is counted once (number_a is less than number_b for V1/V3/V4; the incident-level flag for V2 marks the later incident in each qualifying pair).

Interpretation guide

Outcome	Reading
Pair counts rise steadily 7 to 30 days	Expected; the chosen window should be justified by operational logic, not by the biggest number
Ranking of top caller + category combinations is stable across windows	Confirmed: the two candidate patterns are not an artefact of the window
Excluding Callers 1904 and 290 removes most pairs (87% at 14 days)	Pairs fall from 73,359 to 9,647, but flagged incidents fall only from 25.2% to about 22.8%: repeats persist without them
v2 results differ materially from v1 for backlog and SLA, not for recurrence	Confirmed: DQ-08 mattered for row-selection-dependent fields; recurrence fields are independent of it

F. Data quality register (summary)

ID	Issue	Status
DQ-01	Event log, not an incident table (119,998 rows, 20,769 incidents)	Resolved
DQ-02	Ties on final-row rule (22,136 rows before tie-breaker)	Resolved
DQ-03	Two terminal states plus an artefact state (-100)	Resolved
DQ-04	Day-first text dates; date maths returned NULL	Resolved
DQ-05	? placeholders (asset, group, category)	Mitigated
DQ-06	Asset field unusable for recurrence (99.75% missing)	Mitigated
DQ-07	MIN/MAX on text dates gave impossible spans	Resolved
DQ-08	Final-row ordering used text timestamps	Resolved
DQ-09	Pair explosion (73,359 pairs)	Mitigated
DQ-10	Extreme callers (1904, 290) may be automated or shared accounts	Mitigated
DQ-11	Backlog reflected the DQ-08 row-selection bug, not a real backlog	Resolved
DQ-12	Loaded file smaller than the public dataset description	Cause unconfirmed

Full detail for every item is in *documentation/Data_Quality_Log.md*. DQ-08 changed the selected row for 8,018 of 20,769 incidents (38.6%), moving SLA breach from 16.9% to 39.4% and backlog from 38.6% to ~0%; recurrence, category volume and resolution-time findings are unaffected. DQ-12 is a documented discrepancy, cause not confirmed: the loaded file spans 29 Feb to 13 May 2016 and has the full 36-attribute schema.

G. Reproducibility checklist

- MySQL 8.0 or later (window functions). Python 3.9+ with pandas and NumPy.
- Place the CSV at `data/incident_event_log.csv`.
- Run `sql/01` to `sql/06` in order. Confirm `SELECT COUNT(*) FROM incidents_final_v2` equals 20,769.
- Compare printed results with this appendix; every figure has been run and reconciled.
- Run `python/recurrence_validation.py` with `--final-row-mode text`, `--final-row-mode parsed` and `--exclude-callers` to reproduce sections D and E.